Compact Lie Groups

Based on Lectures given by

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1 Basic Lie Theory

1.1 Lie Groups

Definition 1.1. A Lie group is a pair (G, m) such that G is a smooth manifold and $m : G \times G \to G$ is a smooth map giving G a group structure.

Remark 1.1. We can derive the smoothness of the inversion $G \ni g \overset{i}{\mapsto} g^{-1} \in G$ from Definition 1.1.

Definition 1.2. Let $f : M \to N$ be a smooth map of smooth manifolds. $p \in M$ is said to be regular if $T_p f$ is surjective. A point $q \in N$ is a regular value if $f^{-1}(q)$ is non-empty set consisting of regular points.

Definition 1.3. A subset $N \subseteq M$ of a smooth manifold is called an embedded submanifold if for any point $p \in N$, there is a chart (U, φ) of M such that $\varphi(U \cap N) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$ for some k.

Theorem 1.1. If $q \in N$ is a regular value of $f : M \to N$ then $f^{-1}(q)$ is a closed embedded submanifold of M.

Example 1.1. $(S^1, +)$ is a compact Lie group. Indeed consider

$$\mathbb{R}^2 \ni (x, y) \overset{f}{\mapsto} x^2 + y^2 \in \mathbb{R}.$$

Then for any $(x, y) \in f^{-1}(1)$, $T_{(x,y)} f = (2x, 2y)$. Hence 1 is a regular value of f and in particular, $f^{-1}(1) = S^1$ is a closed embedded submanifold.

Lemma 1.1. A map $\det : \mathbb{R}^{n \times n} \to \mathbb{R}$ is differentiable at any $A \in \mathrm{GL}_n(\mathbb{R})$. Furthermore, the derivative is given explicitly as follows.

$$d[\det](A)(H) = \det(A)\mathrm{tr}(A^{-1}H).$$

Proof.

Base Case : $A = I$.

$$d[\det](I)(H) = \det(I + H) - \det(I) \mod o(H).$$

Write $H = (h_1|\cdots|h_n)$ where $h_1, \cdots, h_n$ are column vectors. Then

$$\begin{aligned}\det(I + H) &= \det(I) + \sum_{i=1}^{n} \det(e_1|\cdots|h_i|\cdots|e_n) \mod o(H) \\ &= \det(I) + \operatorname{tr}(H).\end{aligned}$$

Hence **Base Case** is shown.

General Case. Observe

$$\det(A + H) - \det(A) = \det(A)(\det(I + A^{-1}H) - \det(I)). \tag{1.1}$$

□

Corollary 1.1. 1 is a regular value of det.

Proof. Follows from the non-degeneracy of tr (i.e., $\forall Y \in \mathrm{Mat}_{n\times n}, \operatorname{tr}(XY) = 0 \Rightarrow X = 0$). □

Corollary 1.2. $\mathrm{SL}_n(\mathbb{R})$ is a closed embedded submanifold of $\mathrm{GL}_n(\mathbb{R})$.

Proposition 1.1. For a smooth map $f : M \to N$ and $q \in N$ a regular value, we have

$$T_p f^{-1}(q) = d_p f^{-1}(0) \subseteq T_p M.$$

Definition 1.4.

$$\mathfrak{sl}_n := \operatorname{tr}^{-1}(0).$$

Example 1.2. Let $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$ be a non-degenerate bilinear form. The orthogonal group with respect to $\langle\cdot,\cdot\rangle$ is

$$O(\langle\cdot,\cdot\rangle) := \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \forall x, y, \langle Ax, Ay\rangle = \langle x, y\rangle\}.$$

Notation 1.1. Due to Sylvester's theorem, there is an isomorphism between $\langle\cdot,\cdot\rangle$ and the quadratic form given by the matrix

$$\operatorname{diag}(\underbrace{1, \cdots, 1}_{p}, \underbrace{-1, \cdots, -1}_{q}).$$

With this identification, we may write $O(\langle\cdot,\cdot\rangle) = O(p, q)$.

Remark 1.2. $O(p, q)$ is compact if and only if $p = 0$ or $q = 0$.

Proposition 1.2. For the standard inner product $\langle\cdot,\cdot\rangle$ in $\mathbb{R}^n$, the map

$$\mathrm{Mat}_{n\times n}(\mathbb{R}) \ni A \mapsto A^*\langle\cdot|\cdot\rangle := \langle A\cdot|A\cdot\rangle \in S^2(\mathbb{R}^n)^*$$

is differentiable everywhere and submersive at any $A \in O(p, q)$.

Proof.

$$\begin{aligned}(A+H)^*\langle x|y\rangle - A^*\langle x|y\rangle &\equiv \langle Hx|Ay\rangle + \langle Ax|Hy\rangle + \langle Hx|Hy\rangle \mod o(H)\\ &\equiv \langle Hx|Ay\rangle + \langle Ax|Hy\rangle \mod o(H).\end{aligned}$$

For the surjectivity, we first prove for $A = I$. Since any such symmetric bilinear form is given by $B \in \mathrm{Mat}_{n\times n}(\mathbb{R})$ where B is a symmetric matrix and $\langle Bx|y\rangle$. Using the symmetry of B, we see that by setting $H = \frac{B}{2}$, the surjectivity is shown.

For the general case, set $H' = AH$, then $(H') = H^*A^* = HA^{-1}$ hence the base case can be applied. □

References

[1] Linear algebraic groups.

[2] Linear algebraic groups.